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ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Main)-2016

ANSWERS, HINTS & SOLUTIONS

PART TEST-III
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**D**	**A**	**A**
2.	**D**	**C**	**C**
3.	**A**	**C**	**A**
4.	**B**	**A**	**D**
5.	**B**	**A**	**B**
6.	**A**	**B**	**A**
7.	**A**	**B**	**B**
8.	**C**	**D**	**A**
9.	**C**	**C**	**B**
10.	**A**	**C**	**B**
11.	**C**	**A**	**A**
12.	**B**	**D**	**C**
13.	**C**	**D**	**A**
14.	**A**	**C**	**D**
15.	**C**	**A**	**C**
16.	**D**	**A**	**C**
17.	**D**	**A**	**B**
18.	**A**	**B**	**A**
19.	**A**	**A**	**B**
20.	**A**	**A**	**A**
21.	**A**	**C**	**C**
22.	**A**	**A**	**C**
23.	**B**	**A**	**B**
24.	**C**	**C**	**C**
25.	**B**	**C**	**C**
26.	**D**	**C**	**B**
27.	**D**	**B**	**C**
28.	**B**	**C**	**C**
29.	**A**	**B**	**B**
30.	**D**	**A**	**D**

Physics

PART – I

SECTION – A

1. $4 \times \frac{\mu_0 I}{4\pi}.\left[\sin 45^0 + \sin 45^0\right]\left[\frac{1}{R} - \frac{1}{2R}\right]$

$= \frac{\mu_0 I}{\pi}\left[\sqrt{2}\right]\left[\frac{1}{2R}\right] = \frac{\mu_0 I}{\sqrt{2}\pi R}$ ⊙

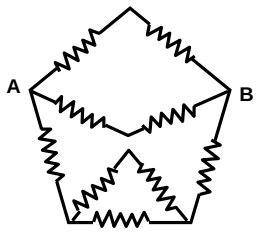
2. $V_1 = \frac{10}{4+4} \times 4$ volts = 5 volt

$V_2 = \frac{10 \times \frac{100 \times 4}{100+4}}{4 + \frac{100 \times 4}{100+4}}$

= 4.9 volts $\therefore \left|V_1 - V_2\right| = 0.1$ volt

3. Equivalent circuit,

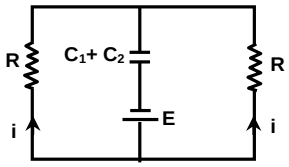
$R_{net} = \frac{8R}{11}$



4. $V_P - V_0 = -\left[10 \times 10 + 10 \times 10\right]$

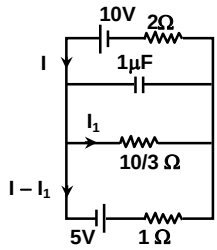
= - 200 volts.

5.



6. $\frac{10}{3}I_1 + 2I - 10 = 0 - 5 + I - I_1 + 2I - 10 = 0$

After solving we get, $I_1 = 0$, Q = 0



7. $V = \frac{1}{4\pi\varepsilon_0 \times 1}\left[q + \frac{q}{2} + \frac{q}{4} + \frac{q}{8} + \ldots\right] = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{1-(1/2)}\right] = \frac{q}{2\pi\varepsilon_0}$

8. Field at any internal point of the cavity.

$$\vec{E} = G\frac{4}{3}\pi\rho\left(\overrightarrow{C_2C_1}\right)$$

Since C_1 and C_2 are fixed (cavity being concentric with C_2 all the time), field remains constant both in magnitude and direction.

9. If the fifth point P has identical charges the field at the centre of symmetry O would be zero. Hence the field due to one single such charge at P would produce a field $\vec{E}'$, such that $(\vec{E}' + \vec{E}) = 0$

$$\therefore \quad \vec{E}' = -\vec{E} = -(-3\hat{i} + 4\hat{j} - 2\hat{k})$$

this field will be four times $\vec{E}'$ due to this rearrangement, hence the required field at O is

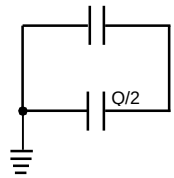
$$4\vec{E}' = (12\hat{i} - 16\hat{j} + 8\hat{k}) \text{ units}$$

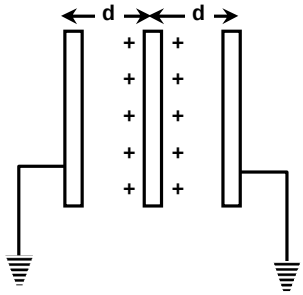
10. If $I_G = 0$ then it becomes a balanced whetstone bridge

$$\ell_1 = \frac{2}{3}\times 1\text{m} = \frac{2}{3}\times 100 = \frac{200}{3}\text{cm}$$

12. $$E_1 = \frac{\sigma}{\epsilon_0} = \frac{Q}{2A\epsilon_0}$$

$$V = E_1 d = \frac{Qd}{2A\epsilon_0}$$





13. Apply gaus theorem

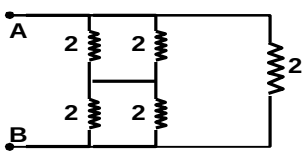
$$E\left(4\pi r^2\right) = \frac{q_{in}}{\epsilon_0} \Rightarrow E = \frac{q}{4\pi\epsilon_0 r^2} \Rightarrow E \propto \frac{1}{r^2} \Rightarrow \text{field depends only on r.}$$

14. $$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\Rightarrow \frac{\Delta R}{R}\times 100 = \frac{\Delta V}{V}\times 100 + \frac{\Delta I}{I}\times 100$$

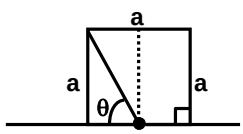
$$= 3\%$$

17. Magnetic field is zero at the centre due to symmetrical current distribution.

18. 

19. $$\text{B at O} = \frac{2\times\mu_0(I/4)}{4\pi(a/2)}\left(0 + \frac{2}{\sqrt{5}}\right) + \frac{\mu_0(I/4)}{4\pi a}\left(\frac{2}{\sqrt{5}} + \frac{2}{\sqrt{5}}\right)$$

$$= \frac{2\mu_0 I}{4\sqrt{5}\pi a} + \frac{\mu_0 I}{4\pi a\sqrt{5}} = \frac{3\mu_0 I}{4\sqrt{5}\pi a}$$



20. Due to electrostatic shielding, the ratio will be 1.

21. When key is open
$q_1 = CE$
when key is closed
$$q_2 = C\left[\frac{E}{R_1 + R_2}R_2\right]$$
$$\therefore \frac{q_1}{q_2} = \left[\frac{CE}{CER_2}(R_1 + R_2)\right] = \frac{(R_2 + R_1)}{R_2} = \frac{5}{3}$$

22. If I_1 is current in ACDB then
$$\frac{\mu_0}{4}\cdot\frac{i_1}{r} = \frac{\mu_0}{4}\cdot\frac{i_2}{4r}$$
$$\frac{i_1}{i_2} = \frac{1}{4}$$
$$i_1 = \frac{1}{5}\times i = \frac{i}{5}$$

23. From symmetry the currents in segment gh is zero. Thus, the equivalent resistance R is given by
$$\frac{1}{R} = \frac{1}{4} + \frac{1}{4} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} \Rightarrow R = \frac{2}{3}\Omega$$
Current in eh = $\frac{10}{4} = 2.5A$
$\therefore$ Potential difference across eh = 5V

24. It is a balanced Wheatstone bridge for capacitor. Consequently the potentials of points a and b are equal. Thus energy stored in the capacitor is zero.

25. Let after steady state the charges on the capacitor be q_1 and q_2
$q_1 = CV$, $q_2 = CV$ where v is the potential difference between the plates.
Also from charge conservation
$q_1 + q_2 = 0$
$\Rightarrow$ V = 0
Thus, final energy stored in the capacitors $= \frac{1}{2}CV^2 + \frac{1}{2}CV^2 = 0$
Initial energy stored = $\frac{q^2}{2C} + \frac{q^2}{2C} = \frac{q^2}{C}$
Thus, heat developed in the resistor = q^2/C

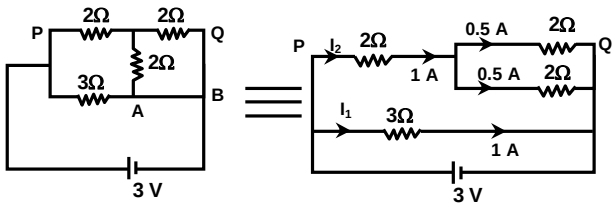
26. Initial capacitor is charged to Q/2
Energy stored in capacitor = $\frac{1}{2}\left(\frac{Q}{2}\right)^2 \times \frac{1}{C} = \frac{1}{8}\frac{Q^2}{C} \Rightarrow \frac{1}{8}CE^2 = U_i$
Finally energy stored in the capacitor, $U_f = \frac{1}{2}CE^2$

Work done by the battery = $\left(\frac{3}{2}CE\right)\times E = \frac{3}{2}CE^2$

$\therefore$ Heat energy lost = $\frac{3}{2}CE^2 - (U_f - U_i) = \frac{3}{2}CE^2 - \left(\frac{1}{2}CE^2 - \frac{1}{8}CE^2\right)$

$= CE^2\left(\frac{3}{2} - \frac{1}{2} + \frac{1}{8}\right) = \frac{9}{8}CE^2$

27. The equivalent circuit is
Current flowing through 3Ω resistance = 3/3 = 1 Amp
Current flowing through P to Q is also 1 amp.
From the figure current flowing through AB is 1 + 0.5 = 1.5 Amp



28. By using the relation

$$B = \frac{\mu_0 i\theta}{4\pi r}$$ where θ is in radian

Here $\theta = 180°$ i.e. π

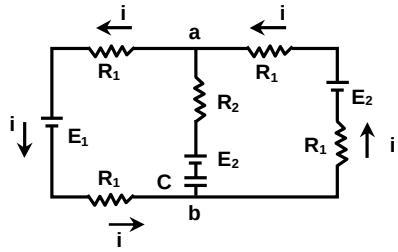
So, $B_p = \frac{\mu_0 I}{4r}$

29. Let x = a′ potential is zero (region II)

$$k\frac{q}{a'} = k\frac{2q}{a + a'}$$

$\Rightarrow$ a′ = a
So, in the region II,
Potential is zero and can be proved.
A solution can also found in region I (x = a/3).

30. In the steady state
$E_2 - iR_1 - iR_1 - E_1 - iR_1 - iR_1 = 0$
$E_2 - E_1 = 4iR_1$
$I = \frac{E_2 - E_2}{4R_1} = (6.3 - 2.1)/(4\times 1.7) = \frac{2.1}{3.4}$
Solving, we get, $V_a - V_b = -iR_1 + E_2 - iR_1$
$= 6.3 - 2\times 1.7\times\frac{2.1}{3.4} = 6.3 - 2.1 = 4.2$ V

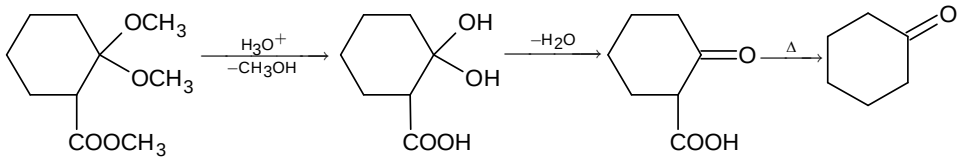


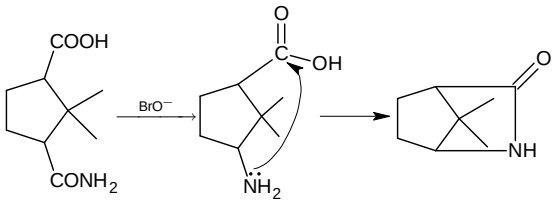
Chemistry

PART – II

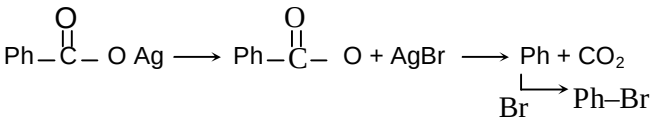
SECTION – A

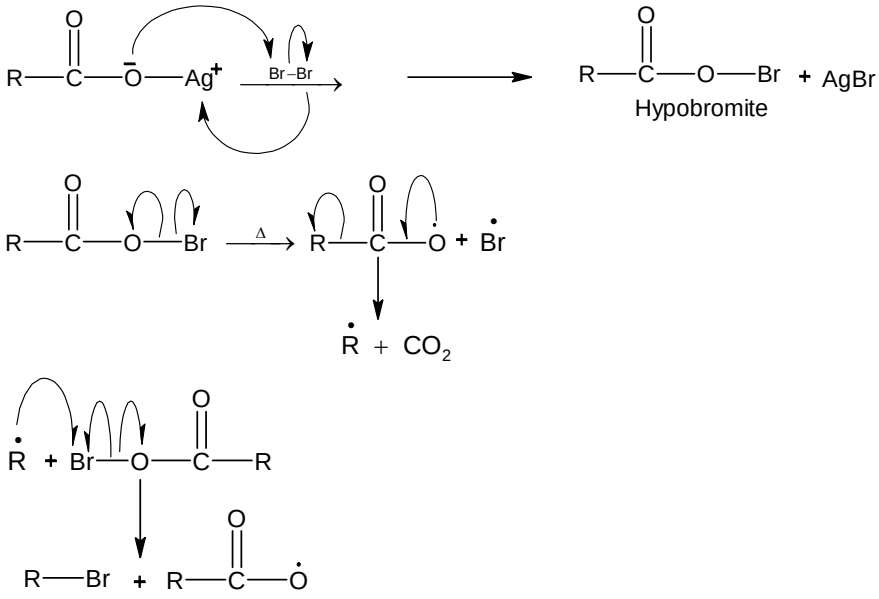
1. MnO_2 oxidized allylic alcohol.

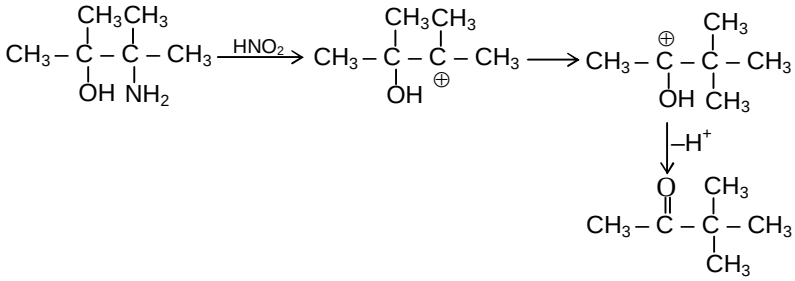
2. 

3. 

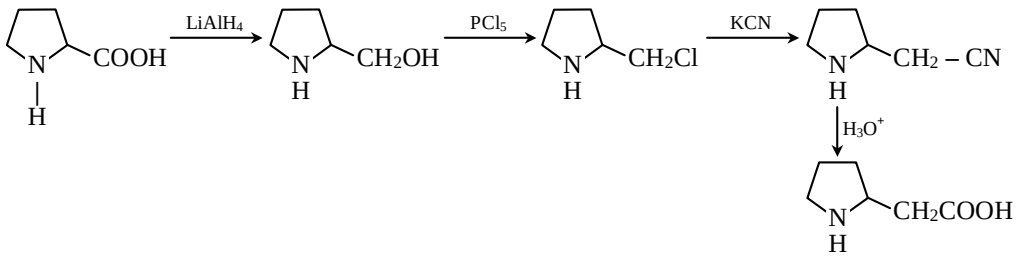
4. This is Hundsdicker reaction.



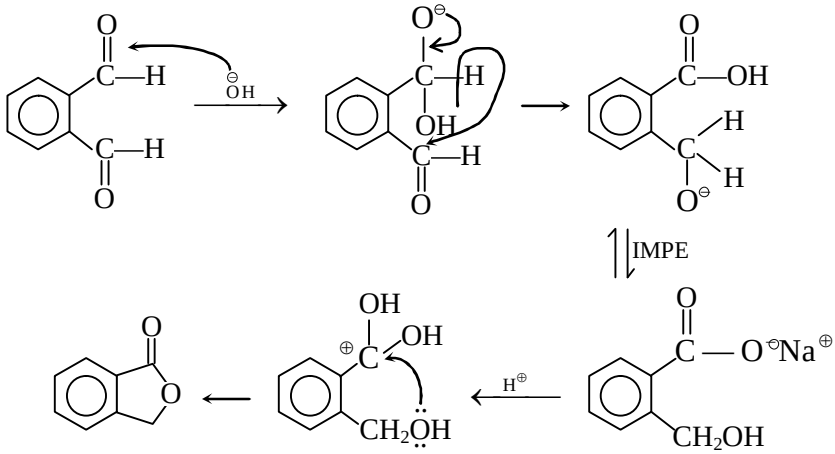
5. 

6. 

8.

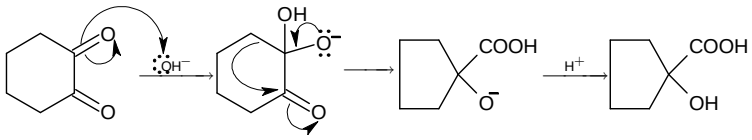


9.

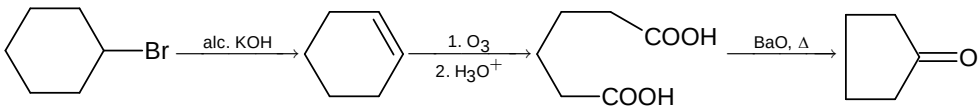


10. Compound (c) does not form enol and hence enolate ion with a base as it does not have $\alpha-$ hydrogen atom.

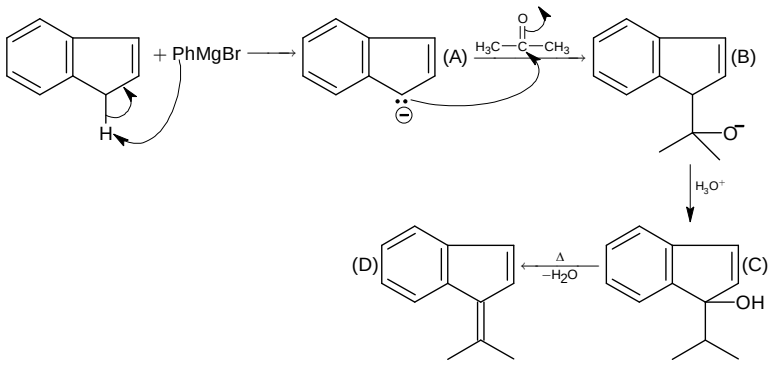
11.



12.



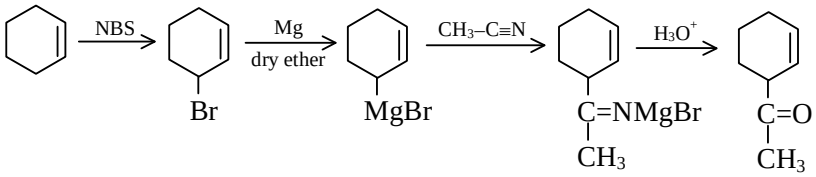
13.

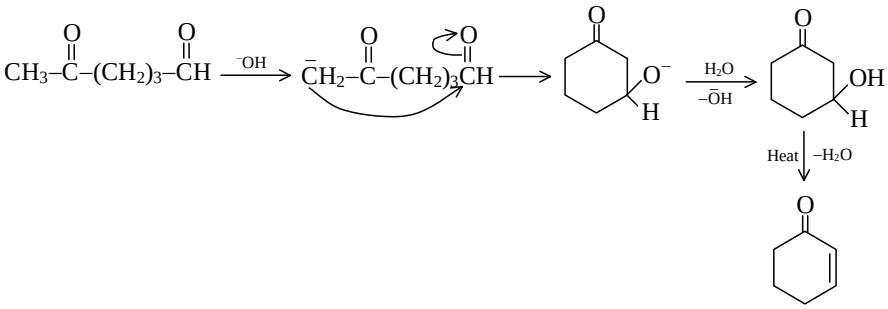


14. Test of terminal alkynes.

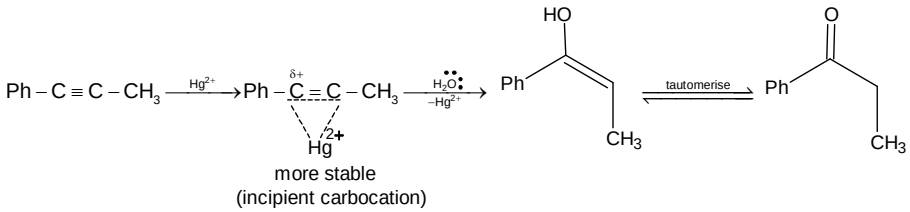
16. In polystyrene the monomeric unit is styrene which is having the formula of $H_2C{=}CH{-}C_6H_5$.

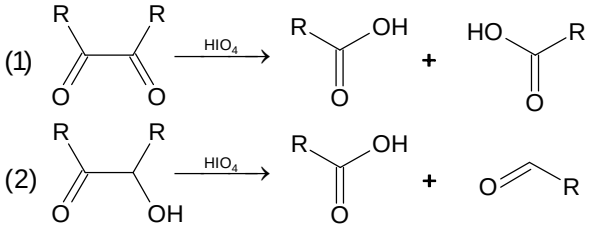
17. $n\ CH_2{=}CH{-}CH{=}CH_2 + n\ C_6H_5{-}CH{=}CH_2$ (styrene) $\rightarrow [CH_2{-}CH{=}CH{-}CH_2{-}CH(C_6H_5){-}CH_2]_n$
(1,3-butadiene) (Buna–S)

18. 

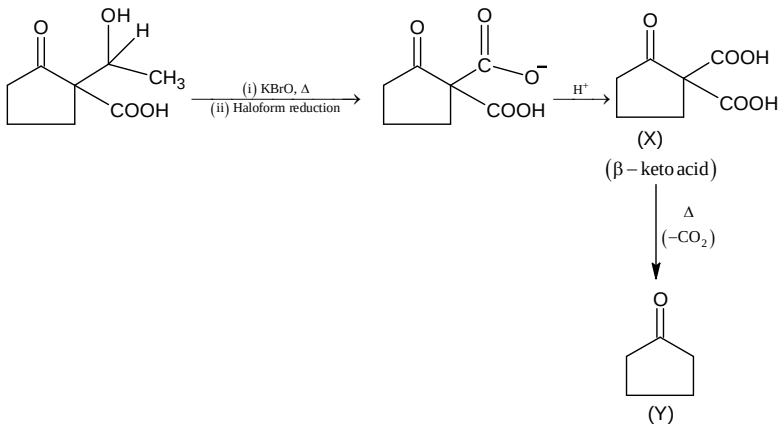
19. 

20. Due to the presence of CS_2, only monobromination takes place as CS_2 decrease the electrophilicity of Br_2 and also minimize phenol ionization.

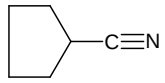
23. 

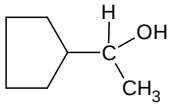
25. 

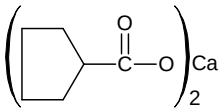
(4) $R-CHOH-CHOH-R \xrightarrow{HIO_4} R-CH=O + R-CH=O$

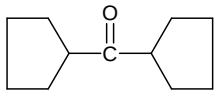
26. 

28.

(A) - 

(B) - 

(C) - 

(D) - 

29. This is an example of Cross – Cannizzaro reaction. The molecule first attacked is converted to sodium salt of benzoic acid in a Cannizzaro reaction. The presence of $-NO_2$ group activates the benzene ring towards nucleophilic attack. Therefore nitrobenzaldehyde is preferentially attacked.

Mathematics PART – III

SECTION – A

1. $\lim\limits_{x\to\infty}\left(\dfrac{3-\sqrt{x}}{2\sqrt{x}+5}\right)\left(\dfrac{\frac{1}{\sqrt{x}}}{\frac{1}{\sqrt{x}}}\right)$

$= \dfrac{\lim\limits_{x\to\infty}\dfrac{\frac{3}{\sqrt{x}}-1}{2+\frac{5}{\sqrt{x}}}}{} = -\dfrac{1}{2}.$

2. Here A & B set having 2 elements is common, so A × B & B × A have 2^2 i.e. four elements in common. Here $n\left[(A\times B)\cap(B\times A)\right] = 4$.

3. $\dfrac{dy}{dx} = \dfrac{r\cos\theta + \sin\theta\dfrac{dr}{d\theta}}{-r\sin\theta + \cos\theta\dfrac{dr}{d\theta}}$

$= \dfrac{(2-2\cos\theta)\cos\theta + \sin\theta(2\sin\theta)}{-2(2-2\cos\theta)\sin\theta + \cos\theta(2\sin\theta)}$

$\left.\dfrac{dy}{dx}\right|_{\theta=\frac{\pi}{4}} = \dfrac{\sqrt{2}}{2-\sqrt{2}}$

4. $f(x) = \log_{\sqrt{2}}\left(2-\log_2\left(16\sin^2 x+1\right)\right)$

$1 \le 16\sin^2 x + 1 \le 17$

$0 \le \log_2(16\sin^2 x + 1) \le \log_2 17$

$2 - \log_2 17 \le 2 - \log_2(16\sin^2 x + 1) \le 2$

now consider

$0 < 2 - \log_2(16\sin^2 x + 1) \le 2$

$-\infty < \log_{\sqrt{2}}\left[2-\log_2\left(16\sin^2 x+1\right)\right] \le \log^2_{\sqrt{2}} = 2$

5. $\left|f\left(g(x)\right)\right| = \begin{cases} 1 & x < -1 \\ 0 & x = -1 \\ -1 & -1 < x < 0 \\ 0 & x = 0 \\ 1 & 0 < x < 1 \\ 0 & x = 1 \\ 1 & x > 1 \end{cases}$

points of discontinuity are x = –1, 0 1.

8. Here f is a differentiable function then f is continuous function so by L.M.V.T. theorem for any a $\in$ (0,4)

$$f'(a) = \frac{f(4) - f(0)}{4 - 0} \quad \ldots (i)$$

again from mean value for any $b \in (0, 4)$

$$f(b) = \frac{f(4) + f(0)}{2} \quad \ldots (ii)$$

multiply both (i) and (ii) we get

$$\frac{f^2(4) - f^2(0)}{8} = f'(a).f(b).$$

9. $f(x) = x^3 + ax^2 + bx + 5\sin^2 x$

$f'(x) = 3x^2 + 2ax + b + 10 \sin x \cos x \geq 0 \ \forall \ x \in R$

i.e. $3x^2 + 2ax + b + 5 \sin 2x \geq 0 \ \forall \ x \in R$

if $3x^2 + 2ax + b - 5 \geq 0$

$4a^2 - 12(b - 5) \leq 0$

$\therefore \quad a^2 - 3b + 15 \leq 0.$

10. Slope of the tangent to curve $y = 2e^x \sin\left(\frac{\pi}{4} - \frac{x}{2}\right)\cos\left(\frac{\pi}{4} - \frac{x}{2}\right)$

$y = e^x \cos x$

$$s = \frac{dy}{dx} = e^x(-\sin x + \cos x)$$

$$\frac{ds}{dx} = e^x(-\sin x + \cos x - \cos x - \sin x)$$

$= -2e^x \sin x$

$$\frac{ds}{dx} = 0 \ \ x = 0, \pi, 2\pi \ (\because \ 0 \leq x \leq 2\pi)$$

value of s at $x = 0$ is 1, at $x = \pi$ is $-e^{\pi}$ at $x = 2\pi$ is $e^{2\pi}$ minimum at $x = \pi$.

11. $$= \int 4\cos\left(x + \frac{\pi}{6}\right)\cos 2x \cdot \cos\left(\frac{5\pi}{6} + x\right)dx = 2\int \left(\cos(2x + \pi) + \cos\frac{2\pi}{3}\right)\cos 2x \ dx$$

$$= 2\int \left(-\cos 2x - \frac{1}{2}\right)\cos 2x \ dx = \int \left(-2\cos^2 2dx - \cos 2x\right)dx$$

$$= -\int (1 + \cos 4x + \cos 2x)dx = -x - \frac{\sin 4x}{4} - \frac{\sin 2x}{2} + c$$

13. Let $y = \sqrt{x + y}$

$y^2 - y - x = 0$

$$y = \frac{1 \pm \sqrt{1 + 4x}}{2} \quad y > 1$$

$$y = \frac{1 + \sqrt{1 + 4x}}{2}$$

14. $$I = 2\int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \cos^2 x} dx$$

$$I = -2\int_0^{\frac{\pi}{2}} \frac{-\sin x}{1 + \cos^2 x} = -2\left[\tan^{-1}(\cos x)\right]_0^{\frac{\pi}{2}}$$

$$= -2\left(\tan^{-1}(0) - \tan^{-1}1\right)$$

$$= \frac{\pi}{2}$$

$$\Rightarrow \quad \frac{\cos\alpha}{1+\sin^2\alpha} = \frac{1}{2}$$

17. $$I = \int \ln\left(\frac{\cos a}{\sin a} + \frac{\sin x}{\cos x}\right)dx$$

$$= \int_0^a \ln\left(\frac{\cos a\cos x + \sin a\sin x}{\sin a\cos x}\right)dx = \int_0^a \ln\left(\frac{\cos(a-x)}{\sin a\cos x}\right)dx$$

$$= \int \ln\cos\{a-(a-x)\}\,dx - \int_0^a \ln(\sin a)\,dx - \int_0^a \ln\cos x\,dx$$

18. $$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k\left[n-(k-1)\right]}{\sum k^2} = \frac{(n+1)\sum_{k=1}^{n}k - \sum_{k=1}^{n}k^2}{\sum k^2}$$

$$= \frac{\frac{n(n+1)^2}{2} - \frac{n(n+1)(2n+1)}{6}}{\frac{n(n+1)(2n+1)}{6}}.$$

19. $$\lim_{x\to0}\frac{(1+x)^{\frac{1}{x}} - e}{x} = \lim_{x\to0}\frac{e^{\ln(1+x)^{\frac{1}{x}}} - e}{x}$$

differentiate two times.

20. $y = e^{-x}\cos x$

$y_1 = -e^{-x}\sin x - e^{-x}\cos x = -e^{-x}\sin x - y$

$y_2 = -e^{-x}\cos x + e^{-x}\sin x - y_1$

$\Rightarrow \quad y_2 = -y - y_1 - e^{-x}\sin x$

$\Rightarrow \quad y_2 = -2(y + y_1)$

$\Rightarrow \quad y_3 = -2(y_1 + y_2)$

$\Rightarrow \quad y_3 = -2(e^{-x}\sin x - y)$

$\Rightarrow \quad y_4 = 2y_1 - 2y - 2y - 2y_1$

$\Rightarrow \quad y_4 + 4y = 0$

25. $$\Rightarrow \quad \lim_{h\to0}\frac{g(a+h)f(a) - g(a)\cdot f(a+h)}{h}$$

$$\Rightarrow \quad \lim_{h\to0}\frac{f(a)\cdot\{g(a+h)-g(a)\} - g(a)\cdot\{f(a+h)-f(a)\}}{h}$$

$\Rightarrow$ f(a) . g′(a) – g(a) . f′(a)

26. $$\lim_{n\to\infty}\left(6^n\right)^{\frac{1}{n}}\cdot\left(1+\left(\frac{5}{6}\right)^n\right)^{\frac{1}{n}}$$

$$\Rightarrow \quad 6\lim_{n\to\infty}\left\{\left[1+\left(\frac{5}{6}\right)^n\right]^{\left(\frac{6}{5}\right)^n}\right\}^{\frac{1}{n}\times\left(\frac{5}{6}\right)^n}$$

$= 6 \cdot e^0 = 6.$

27. For all n it touches the curve.

28. $f(x) = x^3 - 3x + k \Rightarrow f'(x) = 3x^2 - 3$
$f'(x) = 3(x^2 - 1) < 0 \ \forall \ x \in (0, 1)$
Hence $f'(x)$ is strictly decreasing

29. $$y = \tan^{-1}\left(\frac{\log\left(\frac{e}{x^2}\right)}{\log\left(ex^2\right)}\right) + \tan^{-1}\left(\frac{3+2\log x}{1-6\log x}\right)$$

$$= \tan^{-1}\left(\frac{1-2\log x}{1+2\log x}\right) + \tan^{-1}\left(\frac{3+2\log x}{1-3.2\log x}\right)$$

$$= \tan^{-1}1 - \tan^{-1}(2\log x) + \tan^{-1}(3) + \tan^{-1}(2\log x)$$

$= \tan^{-1}(1) - \tan^{-1}(3)$
= constant function.

30. $$\lim_{x\to1}\frac{\sin\left(e^{x-1}-1\right)}{\log x}$$

$$\lim_{x\to1}\frac{\sin\left(e^{x-1}-1\right)}{\left(e^{x-1}-1\right)}\cdot\frac{e^{x-1}-1}{\log x}$$

$$\Rightarrow \quad 1\cdot\lim_{x\to1}\left(\frac{e^{x-1}-1}{\log x}\right) \qquad \left\{\begin{array}{l}\text{as } x\to1\\ e^{x-1}-1\to0\end{array}\right\}$$

$$\Rightarrow \quad \lim_{h\to0}\frac{e^h-1}{h}\times\frac{h}{\log(1+h)} = 1.$$